# Junye Zhou

Portfolio: moonlitjaderabbit.github.io | junye.zhou@mail.utoronto.ca

## Education

**University of Toronto** — Toronto, ON

*Bachelor of Applied Science in Computer Engineering* — *Expected Graduation: 2030*

- Relevant coursework: CIV102 (structures), PHY180 (classical mechanics lab), Praxis I & II (engineering design)

## Additional Experience

**Soma Bone Broth** — Toronto, ON

*Kitchen Hand* — *Jan 2022 – Feb 2022*

- Maintained efficient kitchen operations by preparing ingredients, organizing workstations, and supporting food preparation in a fast-paced environment

## Projects

**Praxis II — Variable Resistance Swim Training System** — Feb 2026 – Apr 2026

*SketchUp 3D, Capstan Design*

- Partner: Mississauga Aquatic Club (MSSAC); RFP focus on variable swim resistance for training
- Designed a capstan-based resistance system providing eight adjustable resistance levels (60–170 N) with approximately 3-second switching time
- Evaluated alternative resistance mechanisms using trade-off analysis and material Pugh charts to balance adjustability, manufacturability, and poolside safety
- Built requirement tables and verification plans; presented at Praxis II showcase

**CIV102 — Matboard Bridge** — Nov 2025 – Jan 2026

*Python, SolidWorks*

- Ranked **#1 in cohort** and **#2 in EngSci Class of 2025** for matboard bridge design and performance
- Designed, analyzed, and fabricated a 1.25 m matboard bridge predicted to withstand 954 N before failure while meeting strict material constraints
- Validated analytical calculations using Python structural analysis to verify predicted bridge performance
- Built and tested full-scale prototype; splice failure near 1000 N drove four design iterations (diaphragms, splice reinforcement)

**PHY180 Damped Pendulum Lab** — Oct. 2025 – Jan. 2026

*OSP Tracker, Python, data analysis*

- Solo lab report: filmed pendulum at 60 fps; tracked motion frame-by-frame in OSP Tracker
- Analyzed period vs. angle, damping, length scaling, and $Q$ vs. length; $Q \approx 109$ (two methods agreed)
- Documented uncertainty sources and compared results to course theory

**Praxis I — Alpha Release Project** — Sept 2025 – Dec 2025

*Proxy testing, Pugh analysis*

- Designed an Alpha Release solution addressing conflicting sleep and study lighting requirements in shared residence rooms
- Diverged on four concepts; ran proxy illuminance tests in a real dorm; converged with Pugh charts
- Recommended foldable bed dome cover targeting <1 lux at the sleeper's eyes

## Technical Skills

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| **Programming:** | Python, Java, C |
| **Engineering Software:** | MATLAB, SolidWorks, OSP Tracker, SketchUp |
| **Engineering Methods:** | Requirements Engineering, Stakeholder Interviews, Pugh Matrix, Concept Selection, Proxy Testing |